

BERN ZOLLIKOFEN, Switzerland

WMO: 066310

Lat: 46.9908N

Lon: 7.4639E

Elev: 1860

StdP: 13.73

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.1	18.9	9.7	9.6	17.8	14.3	12.2	21.3	20.8	48.0	17.5	43.5	4.4	20	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	86.2	67.1	83.0	66.4	79.7	65.1	68.9	82.3	67.5	79.9	66.1	77.5	5.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.3	96.8	73.8	63.0	92.3	72.3	61.9	88.6	71.0	34.1	82.9	32.9	80.0	31.8	78.0	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.8	10.8	DB	9.7	91.3	4.8	3.0	6.3	93.4	3.5	95.1	0.8	96.8	-2.7	98.9
			WB	9.1	71.9	5.0	1.7	5.5	73.2	2.6	74.2	-0.2	75.1	-3.8	76.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	32.8	34.1	41.6	49.3	55.8	63.7	66.7	65.6	57.9	50.5	40.7	33.9
	DBStd	13.78	7.43	7.59	7.02	6.51	6.37	6.46	5.59	5.81	5.83	6.57	6.38	6.73
	HDD50	2226	533	446	268	91	20	1	0	0	7	77	285	498
	HDD65	5927	997	866	725	472	291	101	48	65	220	449	730	963
	CDD50	2034	1	0	8	69	200	411	517	484	246	93	5	0
	CDD65	259	0	0	0	0	6	62	100	83	8	0	0	0
	CDH74	2398	0	0	0	8	89	600	918	716	67	0	0	0
	CDH80	656	0	0	0	0	11	156	271	215	3	0	0	0
Wind	WSAvg	3.9	4.0	4.3	4.7	4.4	4.3	4.2	3.9	3.5	3.5	3.3	3.4	3.8
Precipitation	PrecAvg	46.0	2.5	2.5	2.8	3.5	5.0	5.1	5.3	4.8	4.0	3.9	3.3	3.2
	PrecMax	57.9	6.1	5.8	10.3	7.8	8.9	10.6	11.5	8.5	6.9	6.9	7.1	6.4
	PrecMin	35.6	0.7	0.5	1.1	0.7	1.9	2.1	1.9	0.7	2.0	0.6	0.2	0.2
	PrecStd	6.3	1.5	1.5	1.9	1.7	2.0	2.0	2.0	1.9	1.4	1.5	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	56.6	65.7	75.1	81.0	88.5	90.6	90.8	79.5	70.5	60.6	51.9
		MCWB	48.1	46.4	51.1	57.1	63.4	68.7	68.3	67.6	65.0	58.8	53.0	47.4
	2%	DB	50.1	52.0	61.7	70.3	76.3	84.7	86.2	85.4	75.9	66.9	56.2	49.2
		MCWB	45.2	44.2	49.4	54.9	60.9	67.1	66.6	67.4	63.4	57.8	50.7	45.4
	5%	DB	46.7	48.2	57.9	66.8	72.5	80.9	82.9	81.5	72.0	63.5	52.7	46.5
		MCWB	42.5	42.6	47.6	53.5	58.9	65.7	66.2	66.4	61.4	56.6	48.7	43.5
	10%	DB	43.1	44.7	53.9	62.7	68.5	77.0	79.2	77.4	68.4	60.4	49.9	43.6
		MCWB	40.0	40.3	46.1	51.5	57.2	63.6	64.8	64.8	60.1	55.4	46.6	41.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.9	47.7	52.6	58.2	65.3	71.3	70.3	70.2	66.9	61.3	53.7	48.4
		MCDB	52.9	54.6	62.4	71.9	77.2	85.7	85.8	84.4	76.8	67.3	58.9	51.1
	2%	WB	45.9	44.9	50.4	55.8	62.8	68.6	68.4	68.4	64.5	59.3	51.3	46.0
		MCDB	49.4	50.7	58.9	68.5	73.3	81.2	82.3	81.6	72.9	64.9	55.4	48.7
	5%	WB	42.8	42.9	48.5	53.9	60.3	66.7	67.1	67.1	62.6	57.5	49.3	43.7
		MCDB	46.4	47.9	56.4	65.1	69.3	78.3	79.9	79.1	70.0	62.3	52.4	46.3
	10%	WB	40.2	40.5	46.6	52.1	58.2	64.9	65.7	65.6	60.9	55.7	47.0	41.2
		MCDB	43.0	44.4	53.4	61.3	66.7	75.3	77.0	75.9	67.1	59.8	49.5	43.4
Mean Daily Temperature Range	5% DB	MDBR	10.2	13.0	16.8	18.9	18.0	19.0	19.6	18.6	17.0	14.6	11.3	9.6
		MCDBR	14.2	19.4	24.9	27.0	25.7	25.4	26.3	25.4	22.3	19.7	16.5	12.6
		MCWBR	10.1	12.6	14.3	13.8	12.4	11.1	10.2	10.0	11.4	11.9	11.3	9.5
	5% WB	MCDBR	13.1	18.4	22.1	24.6	22.3	23.5	23.4	22.7	19.8	17.7	15.4	11.6
MCWBR		9.8	12.3	13.0	13.0	11.4	10.8	9.7	9.7	10.6	11.1	10.9	9.1	
Clear-Sky Solar Irradiance	taub		0.290	0.316	0.354	0.392	0.404	0.406	0.406	0.397	0.374	0.350	0.315	0.288
	taud		2.443	2.374	2.320	2.244	2.256	2.294	2.288	2.326	2.373	2.426	2.466	2.451
	Ebn at Noon		256	270	274	273	273	272	271	269	265	254	243	242
	Edh at Noon		21	28	34	41	42	40	40	37	32	26	21	19
All-Sky Solar Radiation	RadAvg		390	658	1032	1382	1591	1884	1828	1590	1228	783	428	316
	RadStd		48	100	154	202	147	176	160	147	112	77	55	55

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A
(47)	Regional (114 neighbors)	N/A	N/A	N/A	+1.35	+0.74	+0.56	N/A	N/A	+103	+29

Nomenclature: See separate page